Unit 10 - Representing Data and Distributions

The following B.E.S.T Standards will be covered in this unit:

MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.

- *Clarification 1: Instruction includes collecting data and discussing ways to collect truthful data to construct graphical representations.*
- *Clarification 2: Within this benchmark, it is the expectation to use appropriate titles, labels, scales, and units when constructing graphical representations.*
- *Clarification 3: Numerical data is limited to positive rational numbers.*

MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation.

- *Clarification 1: Instruction includes choosing the measure of center or measure of variation depending on the scenario.*
- *Clarification 2: The measures of center are limited to mean and median. The measures of variation are limited to range and interquartile range.*
- *Clarification 3: Numerical data is limited to positive rational numbers.*

MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.

- *Clarification 1: Instruction includes recognizing whether a measure of center or measure of variation is appropriate and can be justified based on the given context or the statistical purpose.*
- *Clarification 2: Graphical representations are limited to histograms, line plots, box plots, and stem-and-leaf plots.*
- *Clarification 3: The measure of center is limited to mean and median. The measure of variation is limited to range and interquartile range.*

MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results and draw conclusions about the two populations.

- *Clarification 1: Graphical representations are limited to histograms, line plots, box plots, and stem-and-leaf plots.*
- *Clarification 2: The measure of center is limited to mean and median. The measure of variation is limited to range and interquartile range.*

MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.

Unit 10, Lesson 1: Matching Data Displays

Benchmark

MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.

Learning Target

- ☐ I can match displays of data to the appropriate data set.

10.1.1 Warm-Up: Bell Work

1. During the 2020 NBA Season, Jimmy Butler played in the championship game against the Los Angeles Lakers. The total points scored by Jimmy Butler for the conference semifinals, conference finals, and championship games are listed below.

   $12, 35, 22, 40, 25, 23, 22, 17, 24, 17, 14, 20, 17, 17, 30, 13, 40$

   a. Determine the five-number summary for the data set.

   b. Determine the range.

10.1.2 Exploration: Matching Displays

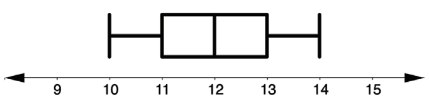
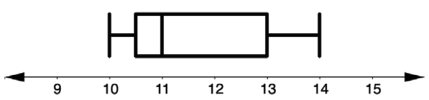
Work with your partner to complete the following.

A middle school band director collects the ages of each clarinet player in the band, shown below.

$$10, 10, 11, 11, 11, 12, 12, 12, 13, 13, 14$$

1. Determine the five-number summary and IQR for the ages of the clarinet players in the band.

2. Which box plot accurately displays the ages of the clarinet players?

Box plot A	Box plot B
 Ages	 Ages

3. Discuss your choice with your partner and summarize.

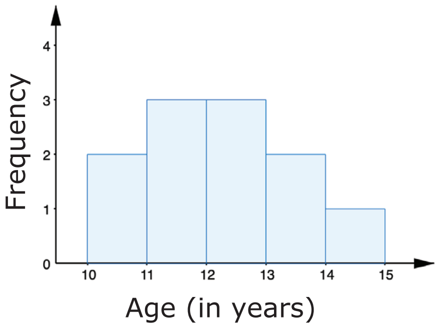
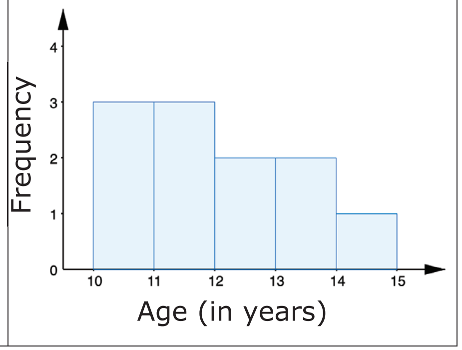
In a box plot, whiskers extend from each quartile to the minimum or maximum. The left whisker represents the bottom 25% of the data and the right whisker represents the top 25%.

4. Discuss with your partner how the location of the median and the length of the whiskers can be used to determine the shape of the distribution. Summarize your discussion.

5. Create a frequency table of the data set.

Age Range	Number of Clarinet Players

6. Which histogram most accurately displays the data?

Histogram A	Histogram B
Age of Clarinet Players 	**Age of Clarinet Players** 

7. Discuss which characteristics of the histogram you used to make your selection.

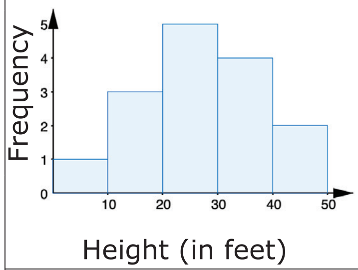
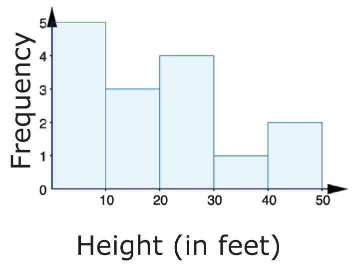
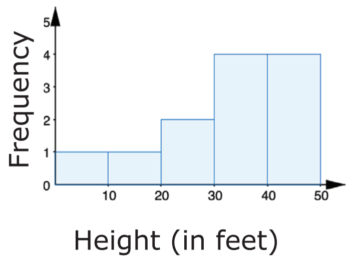
8. Describe the shape of the histogram you chose, including any gaps, outliers, or clusters.

9. Compare the box plot selected for the data with the histogram selected. Discuss which display better represents the data. Summarize your discussion.

10.1.3 Activity: Card Sort

1. An arborist visited a pine tree farm and took measurements of the tree heights, in feet, in three different fields. He dropped his paperwork and all of the data he collected got mixed up.

   Complete the table to match each histogram with its corresponding data set and box plot. Write the number and letter from the card that corresponds to each match.

Histogram	Data Set	Box Plot
		
		
		

2. Compare your answers with your partner. Make any revisions, if needed.

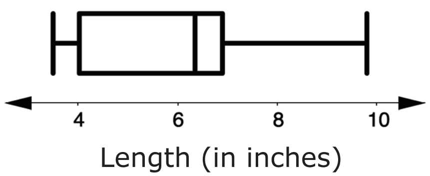
10.1.4 Your Turn!

1. A horticulturist is researching sunflowers to find out the lengths of the petals on a sunflower. She gathers information from 12 sunflowers and measures the length of the petals, in inches, on each flower.

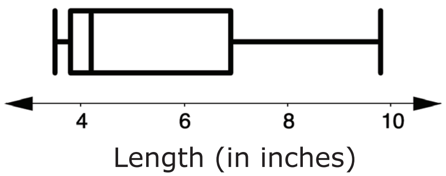
$$6.8, 6.4, 6.2, 3.5, 3.6, 4.1, 3.9, 4.5, 6.7, 7.2, 8.1, 9.8$$

Circle the boxplot and the histogram that best display the data set.

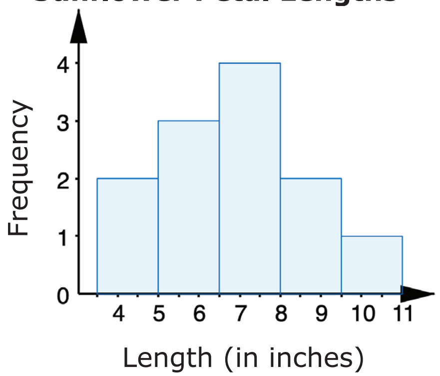
Sunflower Petal Lengths



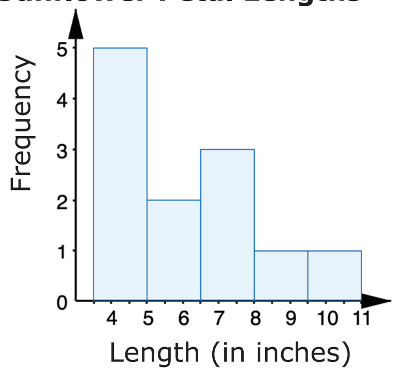
Sunflower Petal Lengths



Sunflower Petal Lengths



Sunflower Petal Lengths



10.1.5 Wrap-Up: Growth Mindset

1. Complete the table as you reflect on today's lesson.

Which mindset did I demonstrate?	Choose the best description.
Did I use the partner discussions as learning opportunities?	○ Never ○ Sometimes ○ Usually ○ Always
Did I work on problems that challenged me?	○ Never ○ Sometimes ○ Usually ○ Always
Did I use strategies to "un-stick" myself when I found the problems difficult?	○ Never ○ Sometimes ○ Usually ○ Always

Unit 10, Lesson 2: Outliers

Benchmarks

MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation.

MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.

Learning Targets

- ☐ I can determine the implications an outlier has in a given data set.
- ☐ I can relate the impact of an outlier to which measures of center and variation best describe a distribution.

10.2.1 Warm-Up: Precision Is Key

Becky Miller is a quality control engineer with GE Aviation. Like many other careers, manufacturing engineering has many moving parts that make up an entire process. Becky says that manufacturing is problem-solving.

1. List at least two steps that you take when faced with a difficult task.

Becky mentions that an error "about the size of a human hair" can make a difference in manufacturing jet engines.

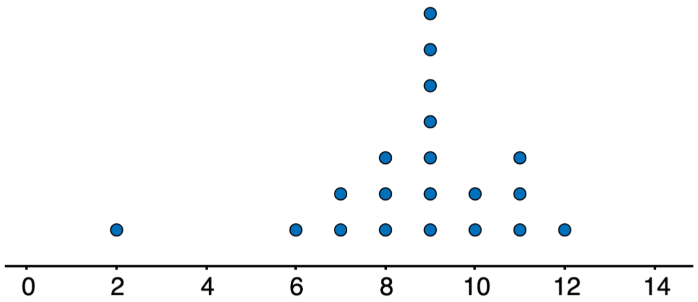
2. What is a time that you have made a small error that caused a big impact in your next steps?

10.2.2 Exploration: Determining, Describing, and Comparing Numerical Data Represented Graphically

Work with your partner to complete the following.

1. Twenty students timed how long it took each of them to solve a puzzle. Below is a list of their solution times, in minutes, and a dot plot displaying the data.

$$2, 6, 7, 7, 8, 8, 8, 9, 9, 9, 9, 9, 9, 9, 10, 10, 11, 11, 11, 12$$



Puzzle Solving Time (in minutes)

   a. Describe the distribution of data.

   b. Determine the mean of the data set. Then, draw an arrow (↑) below the number line that points to this value.

   c. Determine the median of the data set. Then, circle the value on the number line that represents this value.

   d. Discuss with your partner what you notice about the mean and the median of the data. Summarize your discussion.

   e. What is the range of the data?

   f. Find the interquartile range (IQR).

Q1	Q3	IQR

2. Imagine that the fastest time of 2 minutes was removed from the data set. Discuss with your partner what effect, if any, removing this value has on the values of center and variation (mean, median, range, IQR). Summarize your discussion.

3. How does the shape of the graph change after 2 is removed?

4. Compare the new values with the original values.

	Original Value	$>, <, =$	New Value after 2 is Removed
Mean			
Median			
Range			
IQR			

5. Which measure(s) of center will be affected if the outlier is removed?
   - ○ mean
   - ○ median
   - ○ both the mean and median

6. Which measure(s) of variation will be affected if the outlier is removed?
   - ○ range
   - ○ interquartile range
   - ○ both the range and the interquartile range

10.2.3 Bring It Together: Outlier Effects

1. Complete each statement to describe the effect the outlier, 2, has on each measure of center or variation.
   - The presence of an outlier does not have a significant effect on the ○ mean. ○ median.
   - The presence of an outlier will change, often significantly, the value of the ○ mean. ○ median.
   - The presence of an outlier does not have a significant effect on the ○ IQR. ○ range.
   - The presence of an outlier will change, often significantly, the value of the ○ IQR. ○ range.

When outliers are present or a distribution is skewed, the ______________ is the preferred measure of center and the __________ is the preferred measure of variation to describe the distribution.

10.2.4 Guided Practice: Data Outliers

1. Zoe needed exact change to purchase a snack and a drink from the vending machine. She asked her friends how much change, in dollars, they had and listed the results.

   $$0.15, 0.15, 0.20, 0.25, 0.25, 0.30, 0.30, 0.30, 0.35, 0.75, 0.85$$

   a. What do you notice about the data?

   b. Will the mean be higher, lower, or approximately the same as the median? Explain your reasoning or justify your answer by showing your work.

   c. Describe the impact of removing 0.85 from the data set.

   d. The values 0.85 and 0.75 are outliers in this data set. What happens to the range if 0.85 and 0.75 were both removed from the data set?

   e. The IQR of the full data set is __________.

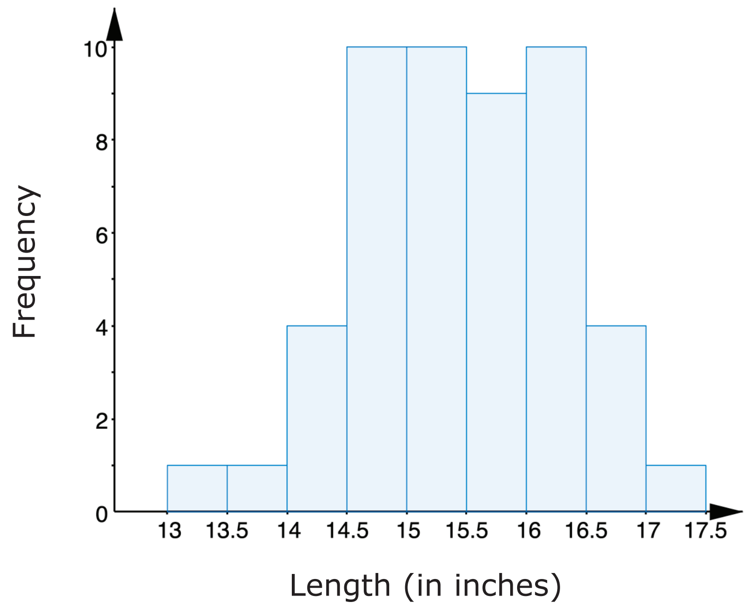
   The IQR with 0.85 and 0.75 removed is ___________.

   f. Is there a big difference between the IQR of the original data set and the modified data set?

10.2.5 Exploration: Measures of Center

1. Consider the histogram of the lengths of female sharks.

Shark Length



a. Describe the shape of the histogram.

b. Circle the value on the horizontal axis of the graph that shows the approximate location of the median of the data.

c. Draw an arrow (↑) on the graph to indicate the approximate location of the mean of the data.

d. Compare your answers to B and C with your partner's answers. Discuss your reasoning and, if necessary, make changes.

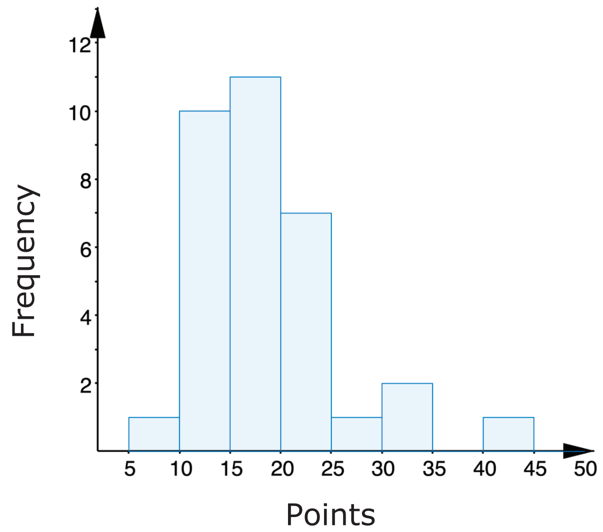
In a symmetrical graph, the mean and the median are located in the middle of the graph, approximately in the same location.

e. The mean and the median of this histogram are located between $15 - 15.5$. Discuss with your partner which measure most accurately describes the center of the histogram. Summarize your discussion.

In a symmetrical graph with no outliers, the ___________ is the preferred measure of center and the ____________ is the preferred measure of variation to describe the distribution.

2. The histogram displays the number of points per game scored by a college basketball player during the 2008 season.

2008 Points per Game



a. Describe the shape of the histogram.

b. Circle the value on the horizontal axis of the graph that shows the approximate location of the median of this data.

c. Draw an arrow (↑) on the graph to indicate the approximate location of the mean of the data.

d. In a symmetrical graph, the mean is the preferred measure of center to use. The histogram for points scored is skewed. Discuss with your partner why the mean would not be a good representation of the center for points scored. Summarize your discussion.

e. In this graph, the median is located between $15-20$ and the mean is in the $20-25$ interval. Which measure most accurately describes the center of the histogram?

f. Which measure of variation should be used to describe the distribution?

In a skewed graph or a graph with outliers, the _____________ is the preferred measure of center and the ______________ is the preferred measure of variation to describe the distribution.

10.2.6 Your Turn!

The following data set shows the number of tickets a movie theater sold in one day for each movie that is playing at the theater.

162	623	76	277	181	64	105
265	82	263	124	167	150	245

1. One of the values in the data set is an outlier. Explain which value you think is the outlier.

2. Complete the statement.

   Removing the outlier, 623, causes the mean to

   ○ increase
   ○ decrease
   ○ remain the same

   and the range to

   ○ increase
   ○ decrease
   ○ remain the same

   because it is much

   ○ larger
   ○ smaller

   than the other values in the data set.

10.2.7 Wrap-Up: Cool Down

1. The presence of an outlier has a significant effect on which measures of center and spread? Select all that apply.

   ☐ Mean
   ☐ Median
   ☐ Interquartile Range
   ☐ Range

2. A data set has an outlier that is significantly less than most of the other values.

   a. Explain how this impacts the measures selected in number 1.

   b. Which measures of center and variation would you use to best describe the data?

Unit 10, Lesson 3: Justifying Measures of Center and Variation

Benchmarks

MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation.

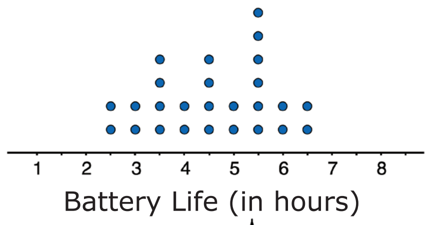
MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.

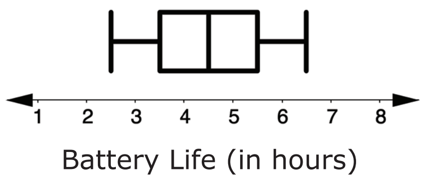
Learning Target

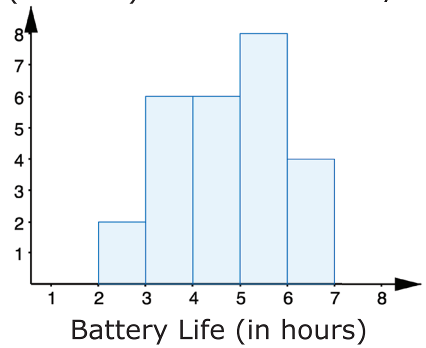
- ☐ I can justify whether or not a measure of center or variation appropriately summarizes data.

10.3.1 Warm-Up: Notice and Wonder

The dot plot, histogram, and box plot summarize the hours of battery life for 26 cell phones that are constantly streaming videos.







1. What do you notice?

2. What do you wonder?

10.3.2 Collaboration: Comparing Data Displays

Suppose an office tracks the number of phone calls received on business days, Monday through Friday, over two weeks. The data set is shown below.

183	165	58	152	179
152	153	135	124	112

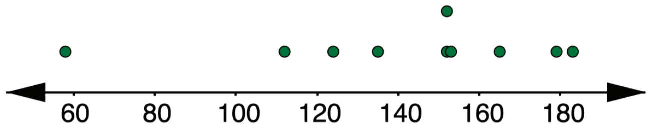
1. Discuss with your partner what you notice about the data set.

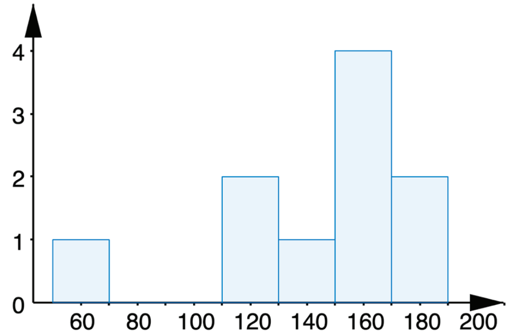
2. Match the measures of center and variation with their calculations for the data set above.

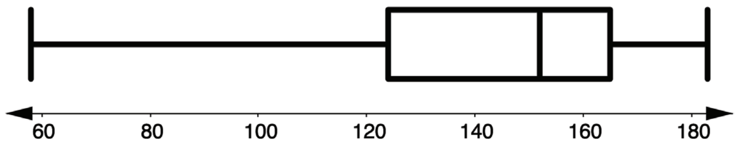
A. Mean	*B. Median*	*C. Range*	*D. IQR*

	$\frac{152+152}{2} = 152$
	$165 - 124 = 41$
	$\frac{183+165+58+152+179+152+153+135+124+112}{10} = 141.3$
	$183 - 58 = 125$

3. A **histogram**, **line plot**, and **box plot** representing the data are shown.







a. Label the display types by filling in the box above each display.

b. Complete the table by identifying which measures of center and variation can be determined from each display.

	Mean	Median	Range	IQR
Histogram	☐	☐	☐	☐
Line Plot	☐	☐	☐	☐
Box Plot	☐	☐	☐	☐

4. Describe any outliers in the data set.

5. Explain which of the displays you used to determine whether or not there was an outlier in the data set.

10.3.3 Guided Instruction: Stem-and-Leaf Plots

Recall the data collected on the number of phone calls an office received on business days over two weeks.

183	165	58	152	179
152	153	135	124	112

In addition to displaying the data graphically with a histogram, line plot, or box plot, the data can also be displayed in a **stem-and-leaf** plot.

A ***stem-and-leaf plot*** is a table that organizes data by place value to compare data frequencies.

1. What do you notice about the data in the stem-and-leaf plot?

Stem	Leaf
5	8
6	
7	
8	
9	
10	
11	2
12	4
13	5
14	
15	2, 2, 3
16	5
17	9
18	3

Key: $5|8 = 58$

2. Compare the stem-and-leaf plot with the line plot from the collaboration.

What similarities do you notice between the displays?	What differences do you notice between the displays?

3. What value does the entry $13|5$ represent?

4. Which measure(s) of center can be determined from the stem-and-leaf plot?

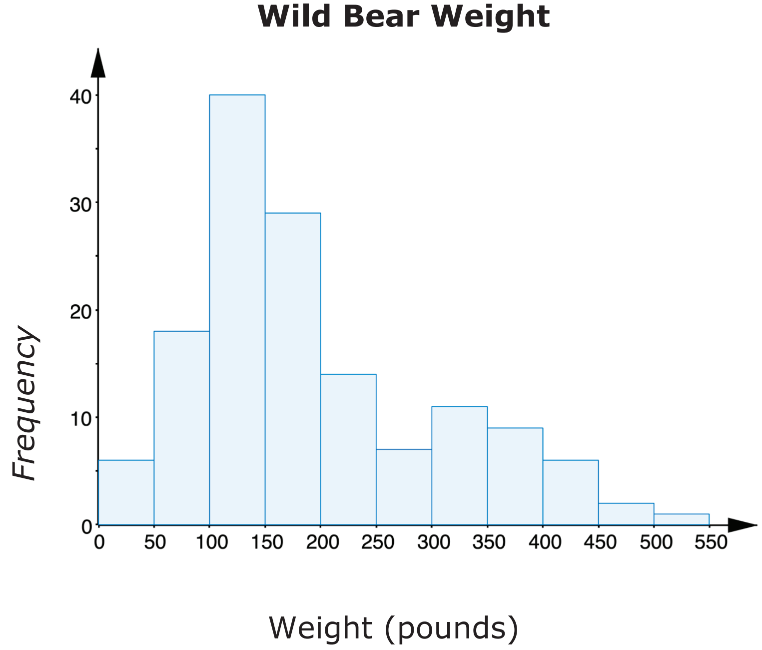
- ○ mean
- ○ median
- ○ both the mean and median

5. Which measure(s) of variation can be determined from the stem-and-leaf plot?

- ○ range
- ○ interquartile range
- ○ both the range and the interquartile range

10.3.4 Collaboration: Center and Variation in Context

1. In a study of wild bears, researchers measured the weights, in pounds (lb), of 143 wild bears that ranged in age from newborn to 15 years old. The data was used to make the histogram shown.



a. Complete the statements.

Based on the data, it appears wild bears typically weigh between

- ○ 50 and 150
- ○ 100 and 200
- ○ 150 and 250

pounds. A good estimate for the typical weight for wild bears is

__________ pounds.

b. Compare your estimate with your partner. Discuss whether it is possible for both of your estimates to be reasonable, even if they are not the same.

c. Discuss with your partner whether your estimate from Part A is closer to the median or the mean. Summarize your discussion.

d. Angelica and Raymond were asked to describe the variation in weight of the wild bears.

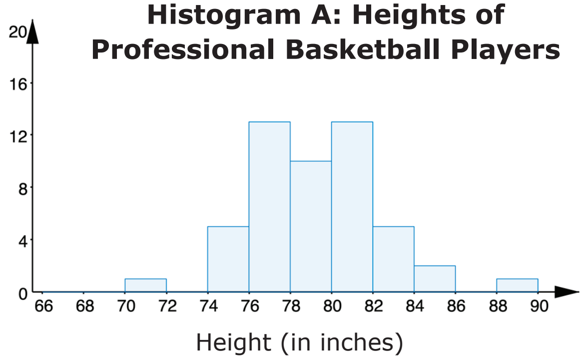
Angelica said, “The bear weights range from less than 50 to over 500 lbs, so the range best describes the spread of the weights of the wild bears.”

Raymond said, “It looks like most bears in the sample weigh between 100 and 300 lbs, with few weighing less and few weighing more. The small weights could be because there were babies in the sample. So, the interquartile range is the best measure of spread to describe this data.”

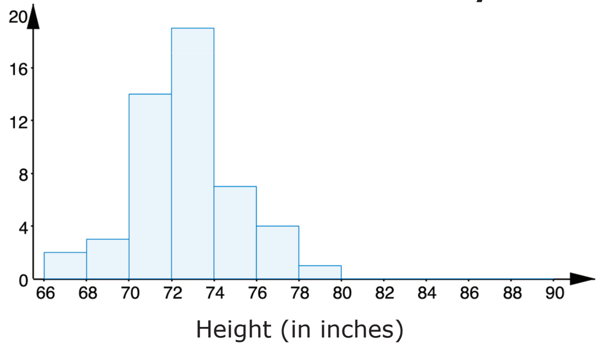
Explain whether you agree with Angelica or Raymond. Justify your reasoning.

2. The two histograms, labeled A and B, show height distributions of 50 male professional basketball players and 50 male professional baseball players.

Histogram A: Heights of Professional Basketball Players



Histogram B: Heights of Professional Baseball Players



Work with your partner to complete each statement.

a. In ○ Histogram A, ○ Histogram B, the average height is around 72 inches. The values of the mean and median are similar.

b. To explain typical height, the variation in ○ Histogram A ○ Histogram B is best described by the interquartile range because the range may be misleading due to outliers. The values of the mean and median are similar.

3. An earthworm farmer set up several containers of a certain species of earthworm so he could learn about their lengths. Lengths of earthworms provide information about their ages. The farmer measured the lengths of the earthworms in one container.

The farmer displayed the earthworms' lengths, in millimeters (mm), in a stem-and-leaf plot.

Stem	Leaf
0	6
1	1, 8, 9
2	0, 3, 3, 5, 5, 6, 7, 7, 8, 9
3	2, 3
4	1, 2, 8
5	2, 4, 9
6	0
7	7
8	
9	3

Key: $2|8 = 28$

Use the display to answer the following questions.

a. Half of the earthworms sampled were less than _________ mm long.

b. The farmer reported that the average length is 35.92 mm. Why might this measure of center be misleading about the typical length of the earthworms in this sample?

c. Compare your answers to Parts A and B with your partner and, if necessary, resolve any differences.

d. The farmer reported that the lengths varied between 6 mm and 93 mm. The farmer's assistant said that most earthworms had lengths between 23 and 48 mm.

Work with your partner to complete the statements.

The farmer is using the ○ IQR ○ range to summarize the variation in worm length, while the farmer's assistant is using the ○ IQR. ○ range.

If a researcher is interested in how long earthworms are, the ○ IQR ○ range is an appropriate measure of variation for the farmer to report because ___________________________
____
____.

10.3.5 Wrap-Up: Reflection

Draw an **X** on the line to indicate where your current learning is in relation to the following learning targets.

1. I can justify whether or not a measure of center appropriately summarizes data.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

2. I can justify whether or not a measure of variation appropriately summarizes data.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

Unit 10, Lesson 4: Determining Appropriate Measures of Center and Variation

Benchmark

MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, while taking into consideration the context and any outliers.

Learning Target

- ☐ I can determine an appropriate measure of center and variation to summarize numerical data.

10.4.1 Warm-Up: Between Two Numbers

The temperature of the sun's core is 15 million degrees Celsius.

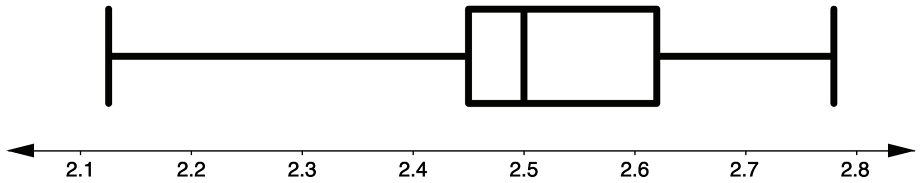
The temperature of Earth's core is 6000 degrees Celsius.

1. What is the ratio of core temperatures of the sun to Earth?

10.4.2 Guided Practice: Measures of Center and Variation

1. The U.S. Energy Information Administration reports weekly average gas prices for the United States. The box plot represents the weekly average gas prices in 2019.

2019 Weekly Gas Prices



Price (in dollars)

   a. Explain which measure of center, the mean or the median, best describes the center of the data.

   b. If possible, find the value of the measure you chose in Part A.

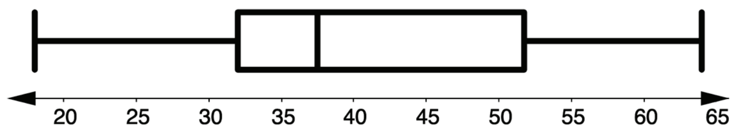
c. Describe the variation of the weekly average price of gasoline.

d. Given your description, does the interquartile range or range better represent the variation in price?

2. Robin, a small business owner, is trying to determine the cost of providing a small group health care plan to her 30 employees. Her insurance agent asks her for a summary of the employees' ages, since age is a factor that affects the premium cost of insurance. The employees' ages are displayed in the stem-and-leaf plot and box plot.

Stem	**Leaf**
1	8, 9
2	5, 8
3	0, 0, 1, 2, 2, 6, 6, 6, 7, 7, 7, 8, 9
4	0, 1, 2, 8
5	1, 2, 5, 8, 8, 9, 9
6	0, 4

Key: $1|8 = 18$



Employee Ages (in years)

Complete the statements to explain which measure of center and variation Robin should provide to her insurance agent to accurately summarize the ages of her employees.

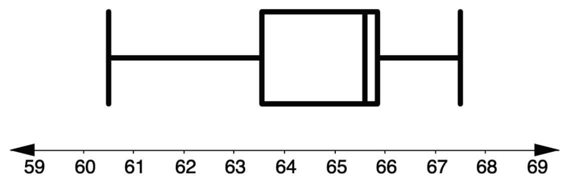
a. The ○ mean ○ median best summarizes the center of the data for the agent because …

b. The ○ IQR ○ range best summarizes the variation of the data for the agent because …

10.4.3 Try It: Measures of Center and Variation

1. Tom Brady is considered by most to be one of the best NFL quarterbacks of all time. His pass-completion percentages from 2009 to 2019 are shown in the box plot and stem-and-leaf plot.

Tom Brady's Completion Percentages 2009-2019



Stem	Leaf
60	5, 8
61	
62	
63	0
64	1, 4
65	6, 7, 8, 9
66	3
67	4

Key: 60|5 = 60.5%

a. The ○ mean ○ median best summarizes the center of the data. The ○ mean ○ median is not the best measure of center because it is influenced by the values ○ 60.5% and 60.8% ○ 66.3% and 67.4% that are far from the rest of the data values.

b. The ○ IQR ○ range best summarizes the variation of the data. The ○ IQR ○ range is not the best measure of variation because it is affected by the value ○ 60.5% ○ 67.4% that is far from the rest of the data.

2. The Rachards family is moving to Gainesville, Florida and want to purchase a home in a specific area. They have a budget of $\$160{,}000$. The prices of homes in the area are shown below, in thousands of dollars.

$$150, 302, 201, 212, 150, 160, 140, 145$$

The Rachards are concerned that the homes may be out of their budget, so they ask the realtor to summarize the home prices.

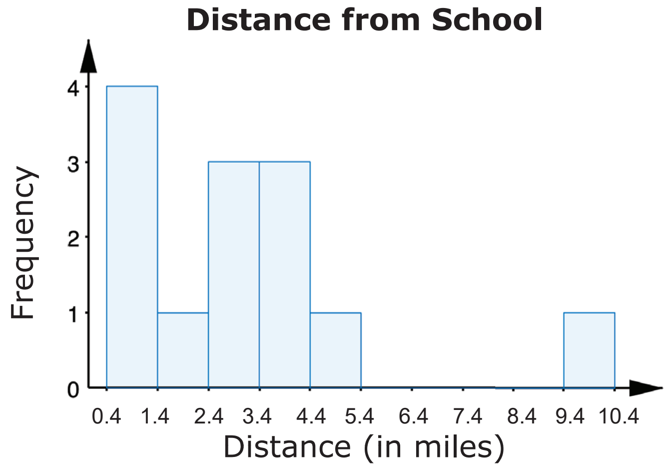
a. Which measure of center should the realtor report to the Rachards, the mean or the median? Explain your reasoning.

b. Which measure of variation should the realtor report to the Rachards, the interquartile range or the range? Explain your reasoning.

10.4.4 Your Turn!

Zane and 12 of his friends listed how far, in miles, they live from the school and created a histogram.

$$0.4, 0.5, 1, 1, 2.4, 2.7, 3, 3.2, 4, 4.2, 4.4, 5, 10.3$$



1. Zane concluded that the best description of the typical distance that a student in his class lives from school is 3 miles, the value of the median.

   Do you agree with Zane? Explain your reasoning.

2. Zane explained to his classmate, Shelby, that the students lived anywhere from 0.4 to 10.3 miles from school, so the range of 9.9 miles was the best summary for the spread of the data.

   Shelby argued, "That's misleading, only one person lived more than 5.4 miles from school, so the interquartile range is a more accurate depiction of the spread of the distances."

   a. Do you agree with Zane or Shelby? Explain your reasoning.

   b. Under what circumstances would you agree with the other person's chosen measure of variation?

3. Noelle is Zane's friend who used to live 10.3 miles away from the school. She recently moved closer, and now only lives 5.5 miles away from the school.

   The data set with Noelle's new distance from school, in green, is shown.

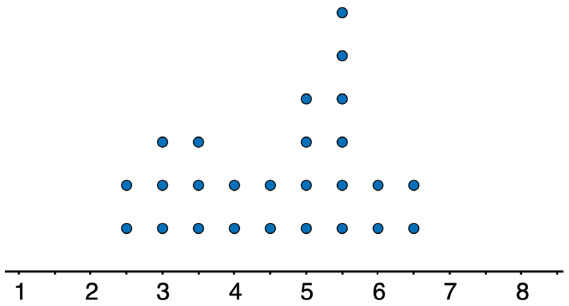
   $$0.4, 0.5, 1, 1, 2.4, 2.7, 3, 3.2, 4, 4.2, 4.4, 5, \mathbf{5.5}$$

   a. What measure of center best describes the typical distance Zane's friends live from school? Explain your reasoning.

   b. What measure of variation best describes the spread of the data distribution? Explain your reasoning.

10.4.5 Wrap-Up: Ticket Out the Door

The dot plot summarizes the hours of battery life for students' cell phones when streaming videos.



Battery Life (in hours)

Mr. Funk's world history class has an upcoming field trip to Tallahassee, Florida that involves a five-hour drive. The students want to be able to watch videos on their phones during the drive without charging their cell phones.

1. Determine the appropriate measure of center and variation for the battery life of the students' cell phones when streaming videos.

Unit 10, Lesson 5: Comparing Graphical Representations

Benchmark

MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.

Learning Target

- ☐ I can use two graphical representations of data to make comparisons and draw conclusions about the populations.

10.5.1 Warm-Up: Defrost Time

1. A microwave shows that it takes 9 minutes and 45 seconds to defrost 1.5 pounds of meat and 26 minutes to defrost 4 pounds of meat.

   a. Without performing any calculations, estimate a range of minutes you think it will take to defrost 3.5 pounds of meat.

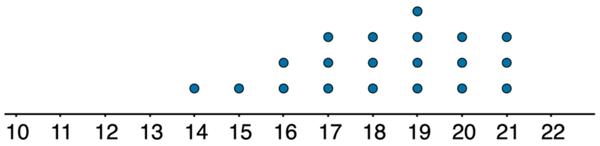
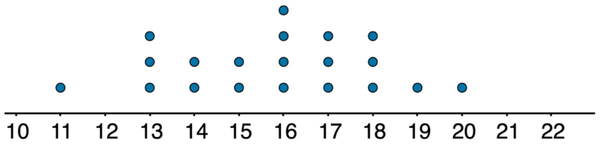
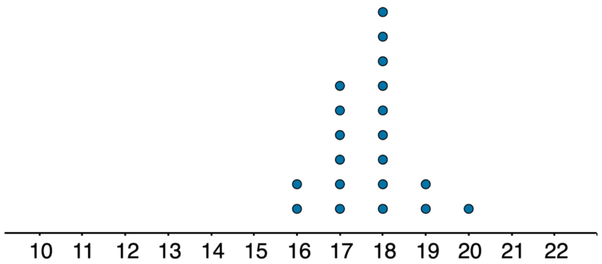
   b. Explain how the given information could be used to find the exact amount of time it takes to defrost 3.5 pounds of meat.

   c. Determine how long it would take for the microwave to defrost 3.5 pounds of meat. Show your work.

10.5.2 Guided Practice: Comparing Line Plots

1. Clare recorded the amount of time a sample of students in the sixth, eighth, and tenth grades spent doing homework, in hours per week. She made a line plot of the data for each grade and provided the following summary:
   - Students in sixth grade tend to spend less time on homework than students in eighth and tenth grades.
   - The homework times for the tenth-grade students are more alike than the homework times for the eighth-grade students.

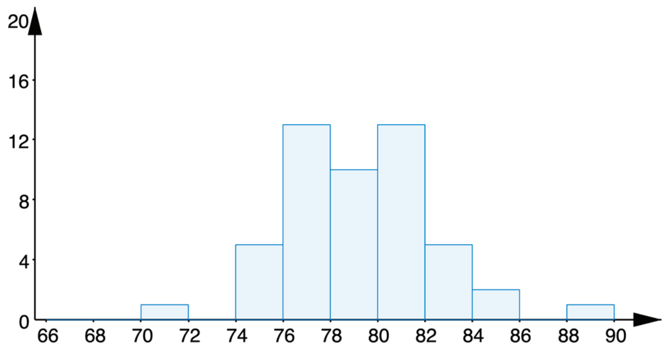
   Use Clare's summary to match each line plot to the correct grade (sixth, eighth, or tenth).

Grade Level	Line Plot
______	
______	
______	 Time (in hours)

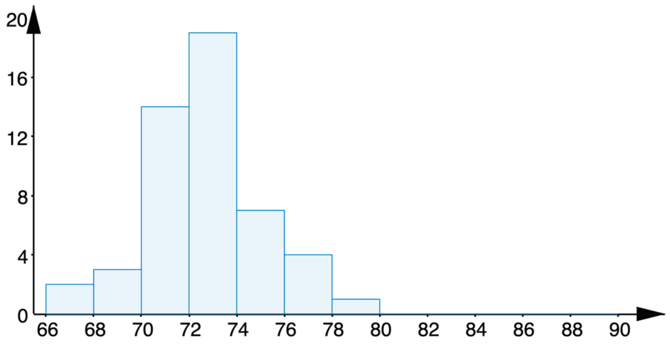
10.5.3 Activity: Station 1 – Histograms and Line Plots

1. The two histograms show height distributions of 50 male professional baseball players and 50 male professional basketball players, in inches (in.).

Heights of Basketball Players



Heights of Baseball Players



a. Complete the statements that draw a conclusion about the typical height of professional basketball players and baseball players.

The sample data in the histograms shows the center of the data for basketball players is

○ less than
○ greater than

the center of the data for baseball players. It can be concluded that professional basketball players are typically

○ shorter than
○ taller than

professional baseball players.

b. Complete the statement.

The histograms show that in the samples, the range of heights of basketball players is

○ less than
○ greater than

the range of the heights of baseball players. We can conclude that professional basketball players have

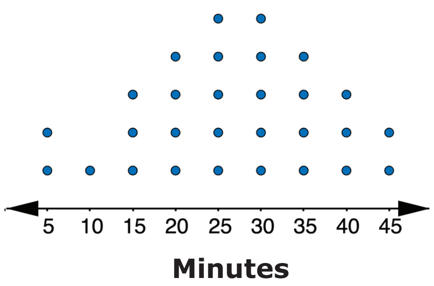
○ more
○ less

variability in their heights than professional baseball players.

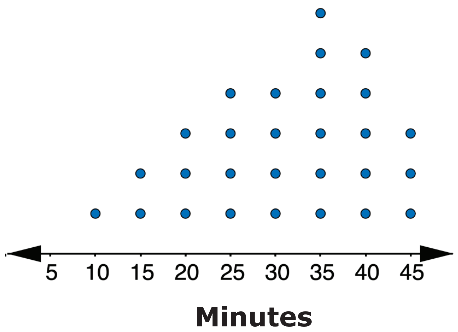
One possible reason for this is …

2. Mr. Logan surveyed a sample of his junior and senior students about the time they spend studying math per day. He then tabulated the results and created a line plot displaying the data for each group.

Time Spent Studying (Juniors)



Time Spent Studying (Seniors)



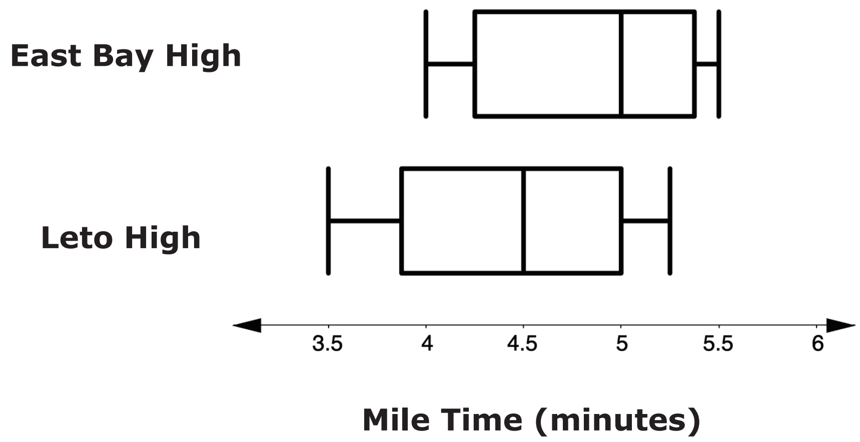
a. The value of the larger median study time for the two groups is _______.

b. The value of the larger mean study time for the two groups is _______.

c. In one to two sentences, describe the difference between the number of minutes Mr. Logan's juniors and seniors study by comparing the center and shapes for the groups.

10.5.4 Activity: Station 2 – Box Plots

1. The mile times for a sample of track team members at East Bay High School and Leto High School are shown in the two box plots.



a. Using the box plots above, calculate and describe each of the features for Leto High in the table.

Feature	East Bay High School	Leto High School
Shape	The shape of the data is asymmetrical.	
Center	The median mile time for the sample of East Bay High team members is 5 minutes.	
Variability	The range of the sample of East Bay High team members' mile times is 1.5 minutes. The IQR of the sample of East Bay High team members' mile times is 0.75 minutes.	

b. Based on the data, explain which team is more likely to win at a track meet.

c. Suppose a runner at a track meet runs a mile in 3 minutes and 45 seconds. Explain which track team you would assume they were on, using information from the data displays.

d. Based on your conclusions and answers to Parts A through C, use some of the terms from the word bank below to write a 4-5 sentence summary comparing the population of runners at East Bay High School to the population of runners at Leto High School.

mean	median	range	IQR
center	variability	upper quartile	
lower quartile	sample	population	

10.5.5 Activity: Station 3 – Stem-and-Leaf Plots and Box Plots

1. A sample of two stores' guitar prices are shown in the stem-and-leaf plots.

Store A's Guitar Prices

Stem	Leaf
2	1, 2, 8
3	3, 6
4	5, 8, 8
5	1, 6
6	
7	2, 2
8	1, 6
9	1

Key: $2|1 = \$210.00$

Store B's Guitar Prices

Stem	Leaf
4	5, 8
5	6, 9
6	1, 1, 3, 5
7	5, 5, 6
8	0, 6
9	1, 8
10	0

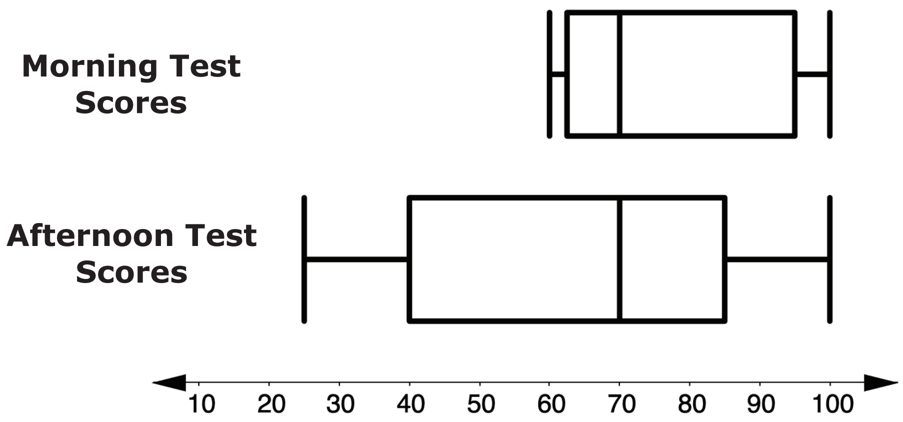
Key: $2|1 = \$210.00$

Zach wants to purchase a guitar and is trying to determine which store to purchase from. Using the displays he makes the following conclusion.

"Store B is better for more experienced guitar players because it typically sells more expensive guitars than Store A. The median price of Store B's guitars is $\$700$, while the median price of Store A's guitars is $\$480$. Prices of guitars at Store A tend to vary more, with an IQR of $\$390$, compared to the prices of guitars at Store B, with a much smaller IQR."

Do you agree or disagree with Zach's conclusion? Explain your reasoning.

2. Suppose a sample of test scores is taken from Mr. Spellman's morning math classes. The data is displayed in the top box plot. Another sample of test scores is taken from Mr. Spellman's afternoon math classes. The data is displayed in the bottom box plot.



Explain the difference between taking Mr. Spellman's math test in the morning compared to the afternoon. Use some of the terms from the word bank and be sure to include both a measure of center and variability in your explanation.

mean	median	range	IQR
center	variability	upper quartile	
lower quartile	sample	population	

10.5.6 Wrap-Up: Reflection

Complete each statement based on your current understanding.

1. I am still unsure about ______________________________
____
____

2. One question I have that will help me better understand is ______________________________
____
______________________________?

Unit 10, Lesson 6: Comparing Data Sets

Benchmark

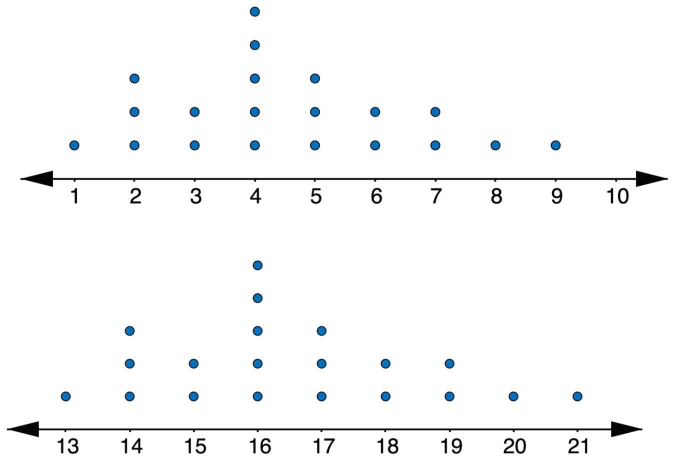
MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.

Learning Target

- ☐ I can compare and interpret two numerical representations of data for two populations.

10.6.1 Warm-Up: Notice and Wonder

The line plots represent the distribution of the tip amounts, in dollars, left at two different restaurants on the same night.



Tip Amount (in dollars)

1. Complete each statement.

   a. When looking at the data displays, one thing I notice is …

   b. One thing I wonder is …

10.6.2 Exploration: Calculating Measures of Center and Variation

On a college entrance exam, the possible scores on the math section range from 200-800. Below are the scores for 20 randomly selected female high school seniors.

454 409 437 415 452 440 454 478 401 403 426 426 437
465 465 429 470 448 511 453

The scores for 20 randomly selected male high school seniors are shown below.

408 440 414 391 480 415 426 392 469 446 409 432 375
388 413 451 370 415 487 404

1. Work with your partner to complete the table by calculating the measures of center and variability for each set of data.

Measure	Female Students	Male Students
Mean		
Median		
Range		
IQR		
Potential Outliers		

2. Complete the statements.

   a. Female students tend to have ○ lower ○ higher scores on the math section of the college entrance exam, with a median of ________, than males, who have a median of ________.

   b. Scores for males tend to have ○ less ○ more variability, with an IQR of ________, than scores for females, with an IQR of ________.

   c. Explain why the measures median and IQR were chosen to describe the data sets above instead of mean and range.

   d. Check your work on Parts A-C with your partner and, if necessary, resolve any differences.

10.6.3 Try It: Drawing Conclusions

Shawn wants to determine the best way to get to school. Over several weeks, he took turns riding in a car and riding his bicycle and recorded the time, in minutes.

Car	
19	17
15	18
12	22
29	14

Bicycle	
24	25
27	26
21	28
32	23

Shawn calculated the measures of center and spread for each transportation method. They are shown in the table.

	Car	Bicycle
Mean	18.25	25.75
Median	17.5	25.5
Range	17.0	11.0
IQR	6.0	4.0

1. Make an argument as to why Shawn should ride in a car to get to school.

2. Make an argument as to why Shawn should ride his bicycle to school.

10.6.4 Activity: Guess My Data Sets

Suppose an English teacher takes five random samples of students and records their most recent test scores.

Data Set A: $\{22, 0, 44, 40, 56, 32, 64, 44, 52, 60\}$

Data Set B: $\{60, 62, 62, 60, 64, 68, 64, 62, 68, 60\}$

Data Set C: $\{70, 82, 74, 76, 80, 72, 84, 70, 78, 82\}$

Data Set D: $\{72, 86, 100, 76, 92, 90, 100, 78, 98, 86\}$

Data Set E: $\{64, 36, 74, 82, 96, 80, 66, 78, 68, 0\}$

Without telling your partner, choose two data sets from above.

1. Write a paragraph comparing the two data sets you have chosen using measures of center and variability. Make sure to refer to the data sets as “Data Set 1” and “Data Set 2” in the paragraph you write.

2. Determine with your partner who is partner A and who is partner B. Write your names on the lines for each partner in the table.

3. Partner A should read their paragraph from Number 1 to partner B, and partner B will guess which two data sets were used. Then, partner B will explain their reasoning for their guess. Both partners will summarize the guess and reasoning in the first row of the table.

4. Then, switch roles so that partner B reads their paragraph while partner A makes a guess and explains their reasoning.

Partner	Guess	Reasoning for Guess
Partner A: ____________		
Partner B: ____________		

5. Reveal to your partner which data sets were used and discuss any errors that may have led to incorrect conclusions.

10.6.5 Your Turn!

1. A company is trying to determine which method of communication is best to tell their employees important information. They send text messages to nine employees and emails to another nine employees. They record how many minutes it takes for the employees to reply.

Text	5	20	45	9	13	12	4	1	11
Email	25	15	35	24	30	19	29	38	12

a. Which measure of center best summarizes the data sets?

Text: ○ mean ○ median Email: ○ mean ○ median

b. Which measure of variability best summarizes the data sets?

Text: ○ range ○ IQR Email: ○ range ○ IQR

c. Calculate the measures of center and variability.

	Text	**Email**
Mean		
Median		
Range		
IQR		

d. Use the most appropriate measures of center and variability to compare the effectiveness of the two methods of communicating with employees.

10.6.6 Wrap-Up: Cool Down

1. Use the data from Your Turn to complete the prompt below.

   Lucas recommends that the company emails its employees because of the smaller range in the data set. Explain why you agree or disagree with Lucas.

Unit 10, Lesson 7: Comparing Data Sets and Graphs

Benchmark

MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.

Learning Target

- ☐ I can compare and interpret a numerical representation of data of one population and a graphical representation of another population.

10.7.1 Warm-Up: Would You Rather?

1. Would you rather…

   A. Pay for 3 gifts and get 1 of equal or lesser value free?
   B. Pay for 4 gifts using a coupon for 20% off?

   Explain your reasoning.

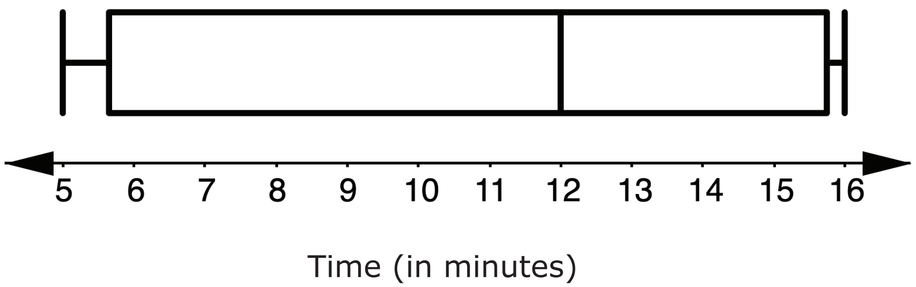
10.7.2 Guided Practice: Summarizing Data

1. The commute time, in minutes, for 24 randomly selected people in Duval County, Florida is shown.

9	10	15	17	18	20	20	21	22	23	24	25
25	25	25	26	26	29	29	29	32	34	38	40

The box plot displays the commute time, in minutes, for 24 randomly selected people in Union County, Florida.

Travel Time in Union County, Florida



a. Explain why it would be beneficial to create a box plot of the Duval County data when comparing these data sets.

b. What are some measures used to describe data that cannot be determined from a box plot?

c. Use at least three terms from the term bank below to write a three-sentence summary comparing travel times to work in Duval County to travel times to work in Union County.

mean	median	range	IQR
center	variability	upper quartile	
lower quartile	sample	population	

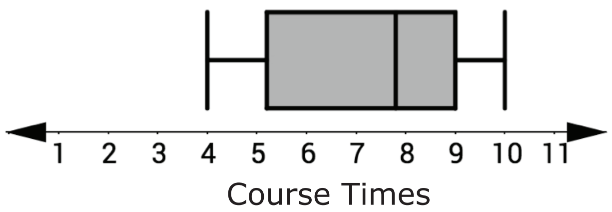
10.7.3 Try It: Comparing Data Sets and Graphs

1. The Deltona High School Wolves and Wildwood High School Wildcats cross-country teams ran an obstacle course. Each team's times are summarized.

Deltona HS Wolves' Obstacle Course Times

4:25	4:43	4:49	5:02	5:12
5:21	5:31	5:32	5:37	5:52
5:54	6:08	6:20	6:26	6:33
6:48	6:53	7:16	7:23	8:05

Wildwood HS Wildcats' Obstacle Course Times



a. Which statements are true about the data for the Deltona Wolves and the Wildwood Wildcats? Select all that apply.

- ☐ The median time of the Deltona Wolves is less than the median time of the Wildwood Wildcats.
- ☐ The fastest 25% of athletes on both teams complete the obstacle course in about the same amount of time.
- ☐ The interquartile range of the Deltona Wolves is less than the interquartile range of the Wildwood Wildcats.
- ☐ Approximately 50% of Wildwood Wildcats have times between 5 and 6 minutes.

b. Which team do you think is more likely to win the next race? Explain your reasoning.

10.7.4 Activity: Roll the Dice

With your partner, decide which of you will be "Partner 1" and "Partner 2." Partner 1 will be creating a line plot and Partner 2 will be recording a data set.

Using a standard six-sided die, each person will roll one die 25 times. After each roll, record the results in either a line plot or numerical data set below, depending on if you are Partner 1 or Partner 2.

1. When both partners are finished, copy their data below.

Line plot (Partner 1's rolls):

Data set (Partner 2's rolls):

2. Complete the table by calculating the measures of center and spread for your data. Share your data with your partner and add their data to the table.

	Partner 1	**Partner 2**
Mean		
Median		
Range		
IQR		

3. With your partner, discuss similarities and differences in your data. Summarize your discussion in the table.

Similarities	**Differences**

10.7.5 Your Turn!

1. A group of ten rowers that practice outdoors on the water (Group 1) are challenging a group of ten who use indoor rowing machines (Group 2). They want to see who can row faster using rowing machines.

   The time, in minutes, it took each of the ten rowers in Group 1 to row 500 meters on the rowing machine is represented in the stem-and-leaf plot.

Group 1: Outdoor Rower Times (in minutes)

Stem	Leaf
15	2, 8
16	8
17	0, 5, 8
18	0, 2, 3, 7

Key: $15|2 = 1.52$

The time, in minutes, it took each of the ten rowers in Group 2 to row 500 meters on the rowing machine is represented in the data set.

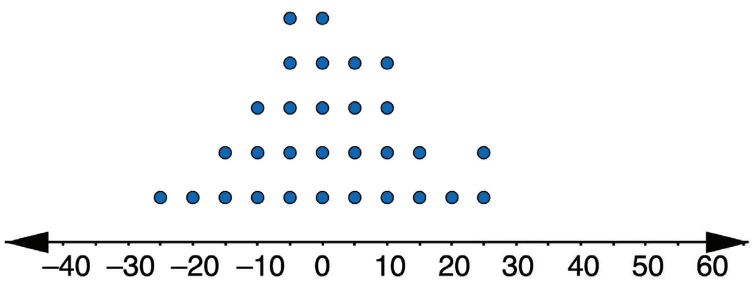
$$\{1.63, 1.82, 1.95, 1.97\ 2.0, 2.02, 2.12, 2.20, 2.25, 2.88\}$$

For each statement, state whether it is true or false. For any statements that are true, explain why.

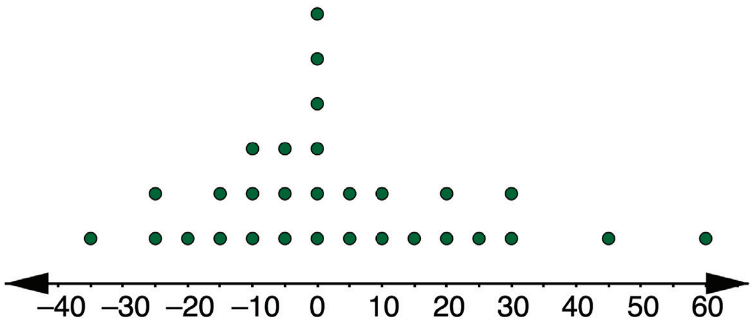
Statement	True or False?	Explanation
The center of the Group 1 times is similar to the center of the Group 2 times.		
The variability of the Group 1 times is similar to the variability of the Group 2 times.		
The Group 1 rowers that practice on the water are typically faster than the Group 2 rowers that row on machines.		
The Group 1 rowers that practice on the water are more consistent than the Group 2 rowers that row on machines.		

2. The dot plots display the arrival times of 30 flights for two different airlines.

Arrival Times of Airline P



Arrival Times of Airline Q



- A negative number represents the number of minutes the flight arrived before its scheduled time.
- A positive number represents the number of minutes the flight arrived after its scheduled time.
- A zero indicates that the flight arrived at its scheduled time.

Based on the data, from which airline would you choose to buy your ticket? Use measure(s) of center and variability to justify your choice.

10.7.6 Wrap-Up: Cool Down

1. Write a short note to a classmate who missed class summarizing today's lesson.

Unit 10, Lesson 8: Random Samples vs. Population

Benchmark

MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.

Learning Target

- ☐ I can determine if a sample is representative of the population.

10.8.1 Warm-Up: Electric Vehicle Technician

Ben Grandy is an electrical vehicle technician for City Car Share. When Ben was considering what to do with his life, he knew he wanted to do something that would bring him happiness. Now he uses the skills he has gained from his study of landscape architecture and work as an electrical engineer to maintain the fleet of electric vehicles and electric bikes used by City Car Share.

Ben found that dedicating himself to a cause that is bigger than himself brings him happiness.

1. Think about what you would like to do with your life. What kind of work would make you happy?

In talking about his work, Ben said, "That's how engineering and design is. You never get to a point where you think, 'Oh, that's perfect! That's done.' You start asking questions like, 'Now that we've gotten this far, what's the next step? How can we improve this?'"

2. How do you think having this mentality is helpful?

10.8.2 Guided Instruction: Is the Sample Representative of the Population?

Many organizations want to know how large groups of people think and behave. This helps businesses make decisions regarding what to sell, when to be open, and whom to target with advertisements. Research is also used to make predictions about events like elections, and to help doctors provide appropriate healthcare to their patients.

To gather this information, companies and organizations often collect data ***samples*** from a ***population*** they are studying.

A ***population*** is the entire set of cases or individuals under consideration in a statistical analysis.

A ***sample*** is a subset, or part, of a population.

1. Imagine a group of researchers want to gather information about which candidate for sheriff all registered voters in Glades County, Florida plan to vote for in an upcoming election.

   Use the definitions to complete the statements.

   In this scenario, the ○ population ○ sample being studied is all registered voters in Glades County, Florida. It is not realistic for researchers to ask every single registered voter, so researchers would collect data from a ○ population ○ sample of registered voters in Glades County, Florida to make predictions.

2. Volusia County School District wants to find out more about which mode of transportation students take to school, so they decide to conduct a survey. The entire school district has a population of $63,223$ students. Since the number of students is so large, the district will only survey some of the students.

   a. In this scenario, what is the population being studied?

   b. What is the sample being used?

To ensure the students surveyed are representative of all students in Volusia County, a system is used to select who is surveyed.

A sample that is **representative** of a population resembles the population and can be used to make accurate predictions about the population.

3. Twenty-five students from each grade level will be chosen at random to participate in the survey. With your partner, discuss how you think this method will help ensure the sample is representative of the population.

4. Claire wants to collect data on the highest eduation level achieved by adults in Jacksonville, Florida. She decides to go to the University of North Florida campus and ask the first person who comes into the library every $\mathbf{30}$ minutes about their education level.

   a. What is the population?

   b. What is the sample?

   c. Explain why a sample chosen in this way may **not** be representative of the population.

A ***random sampling*** is a smaller group of people or objects chosen from a larger group or population by a process giving equal chance of selection to all possible people or objects, and all possible subsets of the same size.

10.8.3 Collaboration: Comparing Methods for Selecting Samples

For each scenario below, read the information about the study and answer the questions that follow.

Scenario 1: Lin is running for president of the seventh grade. She wants to predict her chances of winning the election. She is considering three different methods for surveying a sample of the students.

1. Complete the table to describe the benefits and problems with each of the possible survey methods described.

Sample Surveyed	Benefits of this Method	Problems with this Method
Method 1: Ask everyone on her basketball team who they are voting for.		
Method 2: Ask every third girl waiting in the lunch line who they are voting for.		
Method 3: Ask the first 15 seventh graders to arrive at school one morning who they are voting for.		

2. How could the different methods for selecting a sample in Scenario 1 lead to different conclusions about the population?

3. Explain which of the methods listed in the table would be the most likely to produce a sample that is representative of the population.

4. With your partner, determine a better way to select a sample for this study.

Scenario 2: Maneet, a nutritionist, wants to collect data on how much caffeine the average American drinks per day. She works with three of her colleagues to decide how to survey a sample of the population.

Method 1	Method 2
Maneet and Diarmaid believe the best way to survey is to ask the first 20 adults who arrive at a grocery store after $10:00$ A.M. about the average amount of caffeine they consume each day.	Faith and Santiago would rather ask the first adult who comes into a coffee shop every 30 minutes about the average amount of caffeine they consume each day.

5. What recommendation(s) would you give to this committee about how to select the most representative sample?

10.8.4 Exploration: Sample Size

Isabella is playing a computer game. Her challenge is to create a replica of a hidden spinner that she cannot see. Her score is based on two things: the accuracy of the spinner she creates and the number of spins she uses to create the spinner (the fewer spins she uses equals a higher score).

Isabella creates her first replica after 12 spins of the hidden spinner. Below is a table showing the outcomes of the first 12 spins.

Color	Number of Spins
red	2
purple	4
blue	3
yellow	3

1. Use the data from the first 12 spins to draw a sketch of what you think the hidden spinner might look like.

Isabella's spinner was not accurate, so she had the computer generate additional data. The table below shows the data from the first 36 spins of the spinner, including the 12 shown in the previous table.

Color	Number of Spins
red	3
purple	10
blue	8
yellow	4
orange	2
green	9

2. Revise your spinner based on this new data.

3. Explain which spinner you think is closest to what the hidden spinner looks like.

4. The second spinner Isabella drew was far more accurate than the first. Having additional spins gave a more accurate representation of what the spinner actually looked like. What can this simulation teach us about choosing the size of the sample in a population study?

10.8.5 Your Turn!

1. The meat department manager at a grocery store is worried some of the packages of ground beef labeled as having one pound of meat may be under-filled. He decides to take a sample of 5 packages from a shipment containing 100 packages of ground beef. The packages were numbered as they were put in the box, so each one has a different number between 1 and 100.

   Describe how the manager can select a fair sample of 5 packages.

2. Diego wants to survey a sample of students at his school to learn about the percentage of students who are satisfied with the food in the cafeteria. He decides to go to the cafeteria on a Monday and ask the first 25 students who purchase a lunch at the cafeteria if they are satisfied with the food.
   Explain whether you think this is a good way for Diego to select his sample.

3. Eleanor wants to make the most accurate prediction about who will win the student council election. Explain whether it would be better for her to conduct a random sample of 10 people or 50 people.

10.8.6 Wrap-Up: Cool Down

Anthony is conducting research to determine how students at his school get to school each day (biking, walking, car, bus, etc.). Because there are many students in his school, he will be conducting a survey of a random sample of students to make predictions about the overall population.

1. Select all the options that would produce a random sample to represent the overall population of the school.

   - ☐ Anthony will get to school before everyone else and survey the first 50 people who walk in the doors.
   - ☐ Anthony will survey everyone at his jazz band practice.
   - ☐ Anthony will survey two randomly chosen students in every homeroom.
   - ☐ Anthony will assign each student in the school a number, randomly choose 50 numbers, and survey those students.
   - ☐ Anthony will survey every third student he sees on his walk to school.

Unit 10, Lesson 9: Making Population Predictions – Part 1

Benchmark

MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.

Learning Target

- ☐ I can use proportional relationships to make predictions about a population.

10.9.1 Warm-Up: Notice and Wonder

Burgers	Chips	Fries
Deluxe Burger	\$6.90	\$7.70
Deluxe Cheese	\$7.70	\$8.45
Bacon Burger	\$7.95	\$8.65
Bacon Cheese	\$8.25	\$8.95
Double Burger	\$8.35	\$9.30
Double Cheese	\$9.25	\$9.95
Double Bacon Cheese	\$9.50	\$10.25
Cheese Garden Burger	\$8.25	\$8.85
Mushroom Swiss	\$8.25	\$8.95
Jalapeno Pepper Jack	\$8.25	\$8.95
BBQ Western Bacon	\$9.20	\$9.85
Pastrami Burger	\$9.20	\$9.85
Power Pac	\$10.45	\$11.05
Cheese Power Pac	\$11.25	\$11.70

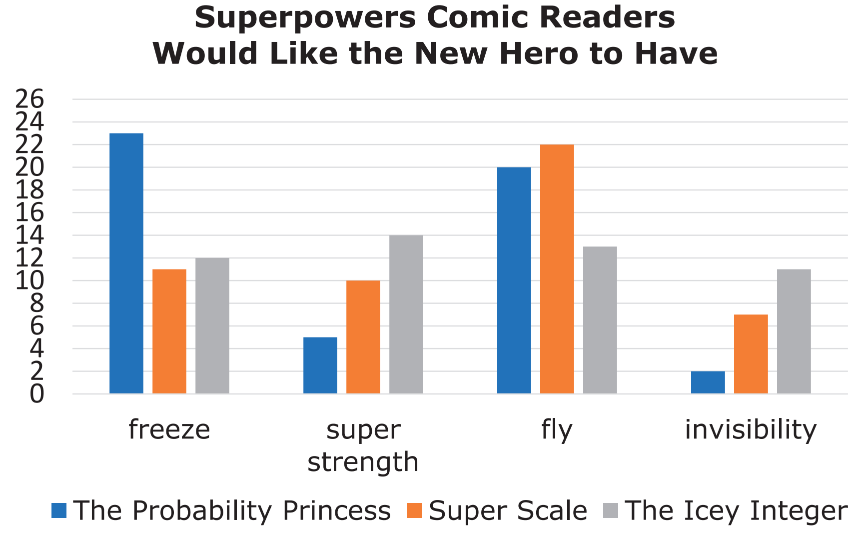
1. What are two things you notice about this menu?

2. What is one thing you wonder?

3. What is a suggestion you could give to the owners about their pricing?

10.9.2 Guided Instruction: Predicting How Many

Results of a random sample can be used to make predictions. The results from a sample of 50 readers of each comic book are shown below.



Suppose the author of *The Probability Princess,* Ashley Grant, is trying to make decisions about the future of the series. She really wants the new hero to have the ability to fly, but she knows there is a larger percentage of readers who want the new hero to have freeze power. As long as the new series can sell at least 3,600 issues in the next five years, Ashley will be able to stay in business.

1. Ashley determines _____ readers of *The Probability Princess* responded to the survey. She then finds the percentage of the 50 readers surveyed who want the new hero to have the ability to fly.

$$\frac{\boxed{}}{50} = \frac{40}{100} = \boxed{} = 40\%$$

Over the past five years, *The Probability Princess* has had $10{,}000$ loyal readers. Using three different strategies, Ashley uses the data to predict the number of those readers who want to see the new hero have the ability to fly. In each method, she predicts that $4{,}000$ of the $10{,}000$ readers want this.

Equivalent Fraction	Scale Factor	Percent
The fraction of readers who want the new hero to fly is $\frac{20}{50}$. Write equivalent fractions until the denominator shows the population total (in this case $10{,}000$). $\frac{20}{50} = \frac{40}{100} =$ $\frac{400}{1{,}000} = \frac{4{,}000}{10{,}000}$	The fraction of readers who want the new hero to fly is $\frac{20}{50}$. Divide the population by the sample to find the scale factor needed. Then, multiply both the numerator and denominator by this scale factor to write an equivalent fraction. $10{,}000 \div 50 = 200$ $\frac{20}{50} \times \frac{200}{200} = \frac{4{,}000}{10{,}000}$	The fraction of readers who want the new hero to fly is $\frac{20}{50}$. Find the percent of the sample for the given condition. Then multiply the population by the percent. $20 \div 50 = 0.4$ or 40% $10{,}000 \times 0.4 = 4{,}000$ OR $40\% = \frac{40}{100} = \frac{2}{5}$ $10{,}000 \times \frac{2}{5} = 4{,}000$

2. Explain whether one of Ashley's strategies seems to be better than the others for solving this particular problem.

3. What is the benefit of solving a problem using two different strategies?

4. Next, Ashley finds the percent of the $10{,}000$ issues they need to sell in the next five years. Complete the work shown below.

$$\frac{3{,}600}{10{,}000} = \frac{\boxed{}}{100} = 0.36 = \boxed{}\%$$

Ashley estimates that if the new hero has the ability to fly, at least $4{,}000$ readers will continue to read and purchase *The Probability Princess*. She estimates that this will lead to each of those $4{,}000$ readers purchasing at least one comic. She also thinks it's likely that readers who did not want the new hero to have the ability to fly may still buy issues in the next five years. Because $4{,}000$ is greater than the $3{,}600$ issues they need to sell to stay in business, Ashley decides to give the next hero the ability to fly.

5. Explain why you agree or disagree with Ashley's decision.

10.9.3 Exploration: Reaction Times

The track coach at Yulee High School in Nassau County, FL needs a student whose reaction time is less than 0.4 seconds to help out with timers at track meets. All the twelfth graders in the school measured their reaction times.

A random sample of 32 students' reaction times has been recorded in the table.

Reaction Times of Less Than 0.4 Seconds	Reaction Times of 0.4 Seconds or More
22	10

1. What fraction of the students in the random sample have a reaction time less than 0.4 seconds?

2. Explain or show how to estimate the percentage of selected twelfth-graders from this school who had a reaction time of less than 0.4 seconds.

3. There are 125 twelfth graders at this school. How many of the twelfth graders are expected to have a reaction time of less than 0.4 seconds?

4. Suppose another group in your class comes up with a different estimate than yours for the previous question. What is another estimate that would be **reasonable**?

5. Explain what kind of an estimate would be considered **unreasonable.**

10.9.4 Guided Practice: Making Population Predictions

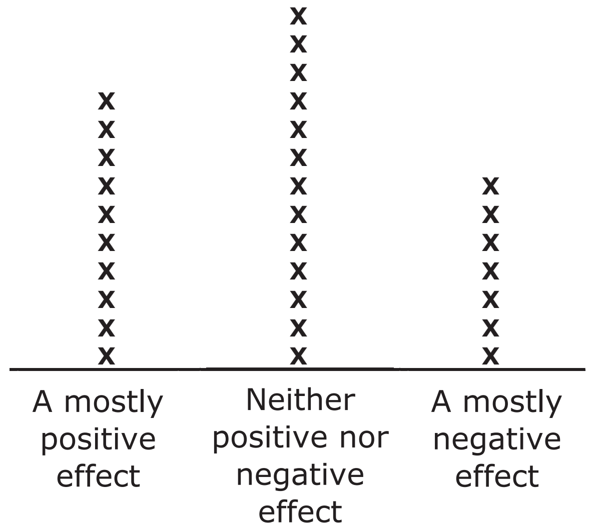
1. Taylor attends Murdock Middle School in Port Charlotte, Florida. She wonders how many students at her school would dye their hair blue if their parents would let them. She surveyed a random sample of 10 students at her school, and 2 of them said they would. Kiran didn't think Taylor's estimate was very accurate, so he surveyed a random sample of 100 students, and 17 of them said they would.

   a. Complete the table below to compare the accuracy of Taylor and Kiran's random samples. Be sure to show your work or explain your thinking.

	Taylor	Kiran
Fraction of students surveyed who would dye their hair blue		
Percentage of students surveyed who would dye their hair blue		
Predicted number of the 500 students in the school who would dye their hair blue		

   b. Explain whose random sample likely produced the most accurate estimate.

2. Ms. Jeffrey teaches seventh grade in Miami, Florida. She noticed many of her students talked about social media and wondered about what her students considered to be the impact of all their online activity. She conducted a random survey of 30 seventh-grade students asking them how they describe the effect of social media use on people their own age. The results are shown in the line plot below.



According to a 2018 Pew Research Center study of American teenagers, 24% of teen respondents said they believed social media has a mostly negative effect on people their own age.

   a. Explain how this compares to the percentage of Ms. Jeffrey's seventh-grade students who described social media as having a mostly negative effect.

There were about 1,546,397 teenagers living in the state of Florida in 2018.

b. Predict the number of teenagers in the state of Florida who considered the effect of social media as mostly positive based on the data Ms. Jeffrey collected.

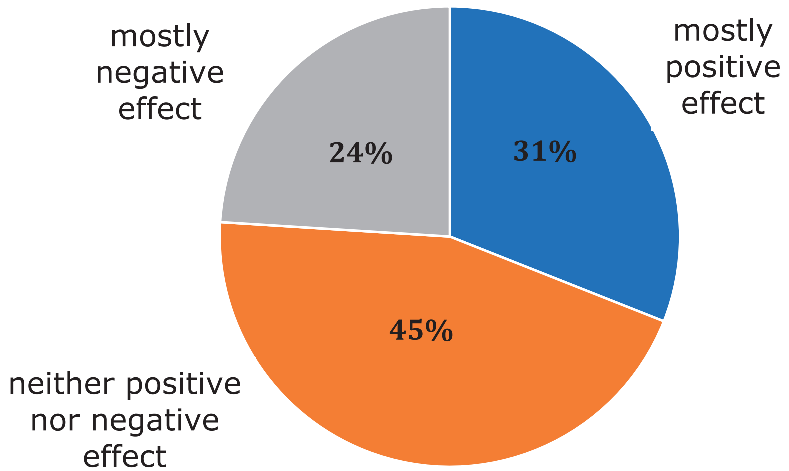
c. Predict the number of teenagers in the state of Florida who considered the effect of social media as neither positive nor negative based on the data Ms. Jeffrey collected.

d. Explain at least two factors that could account for differences in these estimates from the actual number of students who would have described the effect of social media in these ways.

Why are they different?
1.
2.

The results of the 2018 Pew Research Study are shown in the circle graph.

Teen Views on the Effect of Social Media on People Their Age



e. The percentage of teens who describe social media as having a positive effect on people their age is ○ less than ○ equal to ○ greater than the percent of students who responded the same in Ms. Jeffrey's survey.

Write an explanation of why this may have happened.

10.9.5 Wrap-Up: Growth Mindset

Complete the following sentences based on today's learning target:

I can use proportional relationships to make predictions about a population.

1. A barrier to my learning today was …

2. I will try to overcome this by …

Unit 10, Lesson 10: Making Population Predictions – Part 2

Benchmark

MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.

Learning Target

☐ I can use proportional relationships to make predictions about a population.

10.10.1 Warm-Up: Bell Work

Antonia and her aunt go fishing at Lake Okeechobee every Sunday. For the past four Sundays, Antonia has kept track of what kinds of fish they caught.

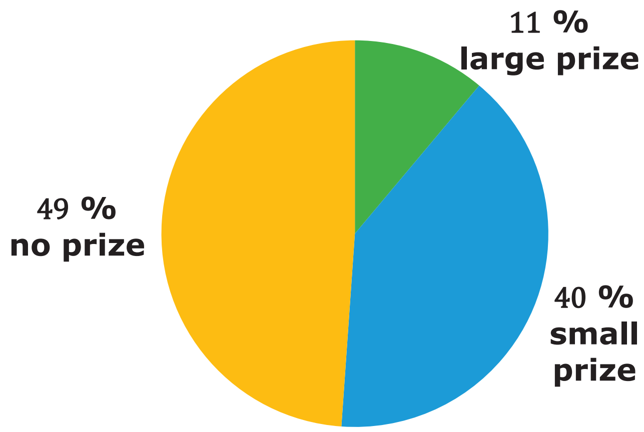
	Bluegill	Bream	Crappie	Largemouth Bass
Number of Fish Caught Last Month	10	12	20	8
Expected Number of Fish to Be Caught Next Sunday				

1. Complete the table to show what type and how many fish Antonia and her aunt are likely to catch next Sunday when they go fishing.

10.10.2 Guided Practice: Making Population Predictions

1. Ella gathered data from a Rubber Duck Pool Game. The circle graph below shows the percentage of people who won a large prize, a small prize, and no prize.

Prizes Won in Rubber Duck Game



There are 180 rubber ducks floating in the pool. The person playing the game takes out one duck and looks for a mark on the bottom of it.

- If there is a blue mark, the person wins a small prize.
- If there is a green mark, the person wins a large prize.
- If there is no mark, the person does not win a prize.

Ella uses her data to predict how many of each kind of duck are in the pool using three different strategies. Her work is shown.

Method 1	Method 2	Method 3
40% of all ducks $\frac{40}{100} = \frac{4}{10} = \frac{72}{180}$ So, 72 ducks have a blue mark.	11% of all ducks $0.11 \times 180 = 19.8$ So, 20 ducks have a green mark.	49% of all ducks $\frac{49}{100} \times \frac{0.56}{0.56} = \frac{27.44}{180}$ So, 27 ducks have no mark.

a. Ella's solution ○ is ○ is not correct because

- ○ the number of ducks with green marks is less than the number of ducks with blue or no marks.
- ○ the number of ducks with no marks is less than the number of ducks with blue marks.
- ○ Ella's solution shows a total of 180 ducks.

b. Write an explanation for why you agree or disagree with Ella's solution. Use specific examples from Ella's strategies to justify your opinion.

c. Compare your explanation with your partner's. Discuss any differences and make changes if necessary.

d. Ella used three different solution strategies. In the bottom of each column in Ella's table, label each strategy with the correct name from the descriptions.

When using a **scale factor,** first divide the population by the sample to find the scale factor, then multiply the numerator and denominator of the sample fraction by the scale factor.

When using an **equivalent fraction,** write multiple fractions equal to the sample fraction until the denominator shows the total population.

When using a **percent,** multiply the total population by the decimal or fraction form of the percent of the sample.

2. Bianca picks blocks out of a bag one at a time. She picks out a block, records the color, and puts the block back in the bag. She cannot see the blocks, but she knows there are 6 blocks in the bag. Of the 50 times she picked a block out of the bag, she picked a blue block 29 times, a red block 10 times, and a yellow block 11 times.

   a. Write ratios to show the number of times Bianca picked a blue, red, and yellow block out of the bag.

   Blue ______ Red ______ Yellow _______

b. Explain which strategy will work best to predict how many of the 6 blocks are blue, red, and yellow.

c. Using the strategy you selected, how many of the 6 blocks are blue, red, and yellow? Show your work.

d. Explain whether the strategy you chose was the best strategy to use, or if another would have worked better.

10.10.3 Collaboration: Stocking a Vending Machine

Students at Sanford Middle School are upset the school vending machine is always sold out of their favorite items. The student council decides to survey students to find a better way to stock the vending machine. They randomly sample 60 students. Each student surveyed picked their 5 favorite items from the vending machine. The results are shown in the table.

Item	Number of Times Chosen in Survey	Number that Should be Stocked
Reese’s	30	
Skittles	24	
Starburst	21	
Snickers	42	
Pretzels	24	
Chips	30	
Takis	39	
Gum	36	
M&Ms	30	
Trail Mix	24	

The vending machine at school has 30 different snack compartments, each with 15 metal spirals for each snack. The vending machine can hold 450 total snacks.

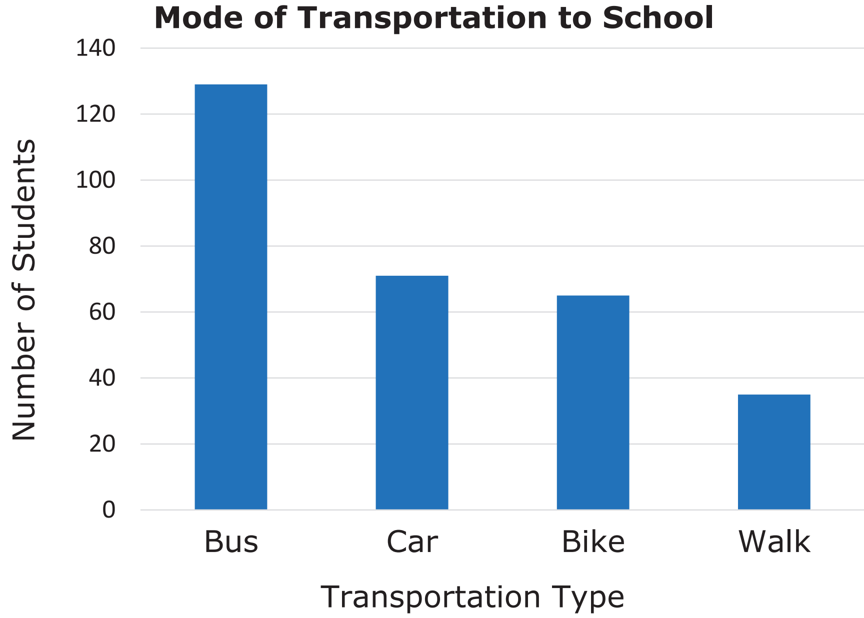
1. Work with your partner to decide how many of each snack should be stocked in the vending machine. Complete the table with your decision.

   Your work will be assessed using these criteria.

   - Each of the strategies: scale factor, equivalent fraction, and percent are used. If one of the strategies is not used, there is an explanation for why the strategy was not used.
   - It is clear, either from your work or a written explanation, why you have stocked the vending machine the way you did.
   - The number of each snack stocked in the vending machine is a direct reflection of the survey results.

10.10.4 Your Turn!

1. Suwannee County School District is studying changes that will need to be made to the transportation department when building new schools. To determine this, it is important to understand how students get to school. The school district has a population of $5{,}959$ students, so a random sample of approximately 5% of the students in each school was conducted. The results of the survey are shown in the bar chart.



   a. Show how to predict the number of students in the district who take the bus to get to school.

b. Show how to predict the number of students in the district who ride their bike to school.

c. Whether or not a student can take the bus to school depends on how close they live to the school. Explain how this data set does not take this into consideration and what effect that might have on the district's decision.

10.10.5 Wrap-Up: Ticket Out the Door

Lin has one month before the seventh-grade class presidential election, and she wants to determine how likely she is to win. Since she can't ask all 665 seventh graders who they are voting for, she conducts a random sample of 105 seventh-grade students. The results of the survey are in the table.

Candidate	Number of People Planning to Vote for the Candidate
Lin	42
Beslinda (Lin's opponent)	30
Undecided	33

1. Use either percent, scale factor, or equivalent fraction to predict how many of the 665 total seventh-grade students currently plan on voting for Lin. Show your work.

2. Use another strategy to predict how many of the total seventh-grade voters are still undecided.